

TECHNICAL SPECIFICATIONS

Al Nakheel Tower Construction Project

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SECTION 1: STRUCTURAL CONCRETE WORKS

1.1 Concrete Mix Design Requirements

All concrete shall be ready-mixed, supplied from approved batching plants, and shall comply with ASTM C94 and ACI 318-19. The following minimum concrete grades shall be used:

Element	Grade	Min. Strength (MPa)	Max W/C Ratio	Min. Cement (kg/m ³)	Exposure Class
Raft Foundation	C50	50	0.35	400	XA2 - Moderate sulfate
Pile Caps	C40	40	0.40	380	XA2 - Moderate sulfate
Basement Walls	C40	40	0.40	380	XA2 + XC3
Core Walls (B3 to L10)	C60	60	0.32	450	XC1
Core Walls (L10 to Roof)	C50	50	0.35	400	XC1
Columns (Ground to L10)	C60	60	0.32	450	XC1
Columns (L10 to L30)	C50	50	0.35	400	XC1
Columns (L30 to Roof)	C40	40	0.40	380	XC1
PT Slabs	C40	40	0.40	380	XC1
Transfer Beams	C60	60	0.32	450	XC1

1.2 Admixtures

All admixtures shall comply with ASTM C494. The following admixtures are specified:

- High-range water-reducing admixture (superplasticizer) for all concrete grades C40 and above, complying with ASTM C494 Type F or G.
- Retarding admixture for concrete placed in ambient temperatures exceeding 35 degrees Celsius, complying with ASTM C494 Type B or D.
- Silica fume shall be added at 8-10% by weight of cement for all concrete grades C50 and above.
- Corrosion inhibitor shall be added to all below-grade concrete elements.

1.3 Reinforcement Steel

All reinforcing steel shall be Grade 60 ($f_y = 420$ MPa) deformed bars conforming to ASTM A615 or ASTM A706. Steel shall be from SABIC or equivalent approved manufacturer. Bar sizes shall range from 10mm to 40mm diameter. All reinforcement shall be cut and bent in accordance with ACI 315 and BS 8666.

Lap splice lengths shall be calculated per ACI 318-19 Chapter 25. Mechanical couplers (Lenton or equivalent) shall be used for bars 25mm diameter and above in congested areas.

1.4 Post-Tensioning System

The post-tensioning system shall be a bonded multi-strand system using 12.7mm diameter 7-wire low-relaxation strands conforming to ASTM A416, Grade 270 ($f_{pu} = 1860$ MPa). The system shall be VSL, Freyssinet, or approved equivalent. Minimum concrete strength at stressing shall be 25 MPa.

SECTION 2: WATERPROOFING

2.1 Below-Grade Waterproofing

A complete tanking system shall be provided to all below-grade structures including raft foundation, basement walls, and pile caps. The system shall comprise:

- a) Primary waterproofing: 4mm thick modified bituminous membrane (e.g., Sika Sarnafil or Carlisle SynTec), torch-applied, with minimum 150mm overlaps.
- b) Protection board: 6mm HDPE dimpled drainage board to protect membrane during backfilling.
- c) Waterstops: PVC waterstops (250mm wide minimum) at all construction joints.
- d) Injection grouting ports at 1.5m centers along all construction joints for remedial injection if required.
- e) Crystalline waterproofing additive (Xypex or equivalent) to be added to all below-grade concrete as secondary protection.

The waterproofing system shall be designed to withstand a hydrostatic head of 15 meters and shall carry a 10-year material warranty from the manufacturer.

2.2 Wet Area Waterproofing

All wet areas (washrooms, kitchens, plant rooms) shall receive liquid-applied polyurethane waterproofing membrane (Sika or BASF), minimum 2mm DFT, with reinforcing fabric at corners and joints. Floor drains shall be waterproofed with preformed collar details. A 48-hour flood test shall be conducted in all wet areas prior to tiling.

SECTION 3: FACADE SYSTEM

3.1 Curtain Wall System

The curtain wall shall be a unitized system with the following performance criteria:

Parameter	Requirement	Test Standard
Air Permeability	Class A4 (max 1.5 m3/h/m2 at 600 Pa)	EN 12153 / ASTM E283
Water Tightness	Class RE 1200 (no leakage at 1200 Pa)	EN 12155 / AAMA 501.1
Wind Load Resistance	Withstand 3.0 kPa positive / 4.5 kPa negative	EN 122179 / ASTM E330
Thermal Performance	U-value max 1.6 W/m2K	EN ISO 10077
Solar Heat Gain Coefficient	SHGC max 0.25	ASHRAE 90.1
Sound Insulation	Min. Rw 40 dB	EN ISO 10140
Seismic Movement	Accommodate +/- 50mm inter-story drift	AAMA 501.4

3.2 Glass Specification

Vision glass shall be insulated glazing units (IGU) comprising: Outer pane: 8mm heat-strengthened low-E coated glass (Guardian SunGuard or equivalent). Cavity: 16mm argon-filled. Inner pane: 8mm tempered clear glass. Total unit thickness: 32mm. All glass above 8 meters height shall be fully tempered to comply with Saudi Building Code requirements for safety glazing.

3.3 Aluminum Framing

All aluminum extrusions shall be 6063-T6 alloy with minimum 3mm wall thickness. Surface finish shall be PVDF (Kynar 500) coating, 2-coat system minimum 50 microns DFT, color to be selected from the architect's palette. All exposed fasteners shall be stainless steel Grade 316.

SECTION 4: MECHANICAL SYSTEMS

4.1 HVAC Design Criteria

Parameter	Office Floors	Lobby/Common	Retail	Basement Parking
Indoor Temp (Summer)	22-24 C	22-24 C	22-24 C	35 C max
Indoor Temp (Winter)	20-22 C	20-22 C	20-22 C	N/A
Relative Humidity	50% +/- 5%	50% +/- 5%	50% +/- 5%	Not controlled
Fresh Air (L/s/person)	10	7.5	10	7.5 L/s/m2
Noise Criteria (NC)	NC 35-40	NC 40-45	NC 40-45	NC 55
Cooling Load (W/m2)	120-150	100-120	150-180	20

4.2 Chilled Water Plant

The central chilled water plant shall comprise 4 x 500-ton (1,758 kW) water-cooled centrifugal chillers (York/Carrier/Trane) with variable speed drives. Chilled water supply temperature: 6.7 C, return temperature: 12.2 C. Condenser water shall be cooled by induced-draft crossflow cooling towers located at roof level. Primary-secondary variable flow pumping arrangement with differential pressure sensors on each floor.

4.3 Fire Protection System

The fire protection system shall comply with NFPA 13 (sprinklers), NFPA 14 (standpipes), NFPA 20 (fire pumps), and Saudi Civil Defense requirements. The fire pump station shall include: 1 x electric motor-driven main pump (150 HP, 2500 GPM at 125 PSI), 1 x diesel engine-driven main pump (identical rating), 1 x electric jockey pump (25 GPM at 140 PSI). Fire water storage: minimum 300,000 gallons dedicated fire tank.

SECTION 5: ELECTRICAL SYSTEMS

5.1 Power Supply

The building shall receive dual incoming 13.8kV power supply from the Saudi Electricity Company (SEC) network. A dedicated 13.8kV ring main unit (RMU) with SF6 gas-insulated switchgear shall be provided. Total estimated connected load: 12 MVA. Maximum demand (with diversity): 8.5 MVA.

5.2 Emergency Power

Two (2) diesel generators rated at 2,000 kVA each (Caterpillar/Cummins/MTU) shall provide emergency power for: all life safety systems (fire pumps, pressurization fans, emergency lighting), elevators (minimum 2 passenger + 1 service), BMS and security systems, and 30% of HVAC capacity. Generators shall be installed in a dedicated room at basement level B3 with fuel storage for minimum 48 hours continuous operation at full load. Automatic Transfer Switches (ATS) shall provide transfer within 10 seconds.

5.3 Lightning Protection

A complete lightning protection system shall be provided in accordance with IEC 62305 and BS EN 62305. The system shall include air termination rods at roof level, down conductors at maximum 20-meter spacing, and a buried ring earth electrode with resistance not exceeding 2 ohms. Surge protection devices (SPDs) shall be installed at all main distribution boards.

5.4 Lighting

All lighting shall be LED with minimum efficacy of 120 lumens/watt. Office areas: 400-500 lux at desk level. Corridors: 150 lux. Parking: 75 lux (50 lux emergency). Lobby: 200-300 lux. A DALI lighting control system shall be provided for all common areas with daylight harvesting, occupancy sensing, and scheduled dimming profiles. Lighting power density shall not exceed 8 W/m² for office floors.

SECTION 6: TESTING, COMMISSIONING & DELIVERABLES

6.1 Testing Requirements

The Contractor shall perform the following testing and commissioning activities:

- a) Concrete cube testing: minimum 6 cubes per 50 m3 of concrete placed, tested at 7 and 28 days. Cores to be taken from structure at rate of 1 per 500 m2 of slab area.
- b) Pile load testing: static load test on 2% of total piles (minimum 5 piles), dynamic testing on 10% of piles.
- c) Waterproofing: 48-hour flood test for all below-grade areas after membrane installation, 48-hour flood test for all wet areas.
- d) Curtain wall: on-site mock-up (3-story, 2-bay) with performance testing for air, water, and structural loads before production.
- e) HVAC: complete TAB (Testing, Adjusting, and Balancing) of all air and water systems by an NEBB-certified agency.
- f) Electrical: thermographic survey of all main distribution equipment under load, power quality analysis for 7-day period.
- g) Fire system: full integrated fire alarm and suppression system test with Civil Defense witness.
- h) Elevator: full load testing, emergency operation, and safety gear testing by third-party inspector.
- i) BMS: 30-day continuous monitoring period with all systems operating under BMS control.

6.2 Required Deliverables

Deliverable	Deadline	Format	Copies
Detailed Construction Program	14 days after Commencement	Primavera P6 + PDF	Digital + 3 hard
Shop Drawings (per package)	21 days before activity start	AutoCAD + PDF	Digital + 5 hard
Material Submittals	28 days before procurement	PDF + physical samples	Digital + 3 hard
Method Statements	14 days before activity start	PDF	Digital + 3 hard
Monthly Progress Reports	5th of each month	PDF + Excel	Digital + 3 hard
Quality Control Reports	Weekly	PDF	Digital + 1 hard
HSE Monthly Report	5th of each month	PDF	Digital + 2 hard
As-Built Drawings	Within 90 days of Completion	AutoCAD + BIM + PDF	Digital + 5 hard
O&M Manuals	Within 60 days of Completion	PDF + hard copy	Digital + 5 hard
Warranties & Guarantees	Upon Substantial Completion	Original documents	3 originals
BIM Model (as-built)	Within 90 days of Completion	Revit/IFC	Digital
Training Records	Before Substantial Completion	PDF + videos	Digital + 2 hard

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